



A Level • Cambridge (CIE) • Chemistry

54 mins 29 questions

Multiple Choice Practice

Topical Practice MCQ

2. Atoms, molecules and stoichiometry

Carefully selected past paper questions by topic.

- 37** An aqueous solution contains 4.00 g of a carboxylic acid, Q. When this solution reacts with an excess of magnesium, 380 cm³ of gas is produced, measured at s.t.p.

What is the relative formula mass of Q?

- A** 59 **B** 118 **C** 126 **D** 236

- 3 Aluminium carbide, Al_4C_3 , reacts readily with aqueous sodium hydroxide. The two products of the reaction are $NaAlO_2$ and a hydrocarbon. Water molecules are also involved as reactants.

What is the formula of the hydrocarbon?



Question 3

9701_w23_qp_13 • Q3

- 3 Aluminium carbide, Al_4C_3 , reacts readily with aqueous sodium hydroxide. The two products of the reaction are $NaAlO_2$ and a hydrocarbon. Water molecules are also involved as reactants.

What is the formula of the hydrocarbon?

A CH_4

B C_2H_6

C C_3H_8

D C_6H_{12}

14 Which statement about atoms and molecules is correct?

- A** The molecular formula of a compound is the simplest whole number ratio of atoms of each element in the compound.
- B** One mole of any substance contains 6×10^{23} atoms.
- C** The relative atomic mass of an element is the ratio of the average mass of one atom of the element to the mass of an atom of carbon-12.
- D** The relative formula mass of a compound is the sum of the individual atomic masses of all the atoms in the formula.

- 3** In separate experiments, 5.0 g samples of each of four s-block metals are added to an excess of water. The gas evolved is collected and its volume measured under the same conditions of temperature and pressure for each sample.

Which metal produces the largest volume of gas?

- A** calcium
- B** potassium
- C** rubidium
- D** strontium

12 A student adds 3 mol of acidified $\text{K}_2\text{Cr}_2\text{O}_7$ to an excess of I^- ions.

The chromium is all reduced to Cr^{3+} and I^- ions are oxidised to I_2 .

The I_2 released is reduced back to I^- ions by X mol of $\text{S}_2\text{O}_3^{2-}$ ions.

1 mol of I_2 is reduced by 2 mol of $\text{S}_2\text{O}_3^{2-}$ ions.

What is the value of X?

A 3

B 6

C 9

D 18

-
- 1 Which sample contains the same number of the named species as the number of molecules in 35.5 g of chlorine?
- A atoms in 16 g of sulfur
 - B atoms in 23 g of sodium
 - C ions in 74.5 g of potassium chloride
 - D molecules in 88 g of carbon dioxide

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- 3** Substance Q is a hydrocarbon. When 1.00 g of Q is completely burned, 3.22 g of carbon dioxide is produced.

What could be the identity of Q?

- A** cyclohexene
- B** cyclopentane
- C** ethene
- D** pentane

- 24** A mixture of ethanol and methanol is burned in oxygen to produce 35 cm^3 of CO_2 and 55 cm^3 of H_2O .

Complete combustion occurs and the volumes of both products are measured at 101 kPa and 120°C .

What is the molar ratio, ethanol : methanol, in the mixture?

- A** 1:3 **B** 2:3 **C** 3:2 **D** 3:1

1 Which statement is correct?

- A Cl has a relative isotopic mass of 35.5.
- B Cl_2 has a relative molecular mass of 70.
- C ICl has a relative molecular mass of 162.4.
- D NaCl has a relative molecular mass of 58.5.

- 2** A 3.7 g sample of copper(II) carbonate is added to 25 cm³ of 2.0 mol dm⁻³ hydrochloric acid.

Which volume of gas is produced under room conditions?

- A** 0.60 dm³ **B** 0.72 dm³ **C** 1.20 dm³ **D** 2.40 dm³

- 2 In this question it should be assumed that $(\text{NH}_4)_2\text{CO}_3 \cdot \text{H}_2\text{O}(\text{s})$ dissolves in water without causing an increase in volume.

Which mass of $(\text{NH}_4)_2\text{CO}_3 \cdot \text{H}_2\text{O}(\text{s})$ should be added to 800 cm^3 of water to form a $0.100 \text{ mol dm}^{-3}$ solution of NH_4^+ ions?

- A 4.56 g B 7.13 g C 9.12 g D 14.3 g

Question 14

9701_w19_qp_13 • Q2

- 2 Diamond is a pure form of carbon. The mass of a diamond can be measured in carats. One carat is 0.200 g of carbon.

Which expression gives the number of carats that contain 6.02×10^{23} carbon atoms?

A 0.200×12.0

B $\frac{0.200}{12.0}$

C $\frac{12.0}{0.200}$

D $\frac{0.200}{6.02 \times 10^{23}} \times 12.0$

- 3** Which fuel would produce the largest mass of CO_2 when 10 kg of the fuel undergo complete combustion?
- A** biodiesel, $\text{C}_{17}\text{H}_{34}\text{O}_2$
 - B** ethanol, $\text{C}_2\text{H}_6\text{O}$
 - C** octane, C_8H_{18}
 - D** propane, C_3H_8

- 28** A sample of 2.30g of ethanol was mixed with an excess of aqueous acidified potassium dichromate(VI). The reaction mixture was then boiled under reflux for one hour. The required organic product was then collected by distillation. The yield of product was 60.0%.

Which mass of product was collected?

- A** 1.32 g **B** 1.38 g **C** 1.80 g **D** 3.00 g

- 2** Which would contain 9.03×10^{23} oxygen atoms?
- A** 0.25 mol aluminium oxide
 - B** 0.75 mol sulfur dioxide
 - C** 1.5 mol sulfur trioxide
 - D** 3.0 mol water

- 24** A sample of 2.76 g of ethanol was mixed with an excess of aqueous acidified potassium dichromate(VI). The reaction mixture was then boiled under reflux for one hour. The required organic product was then collected by distillation.

The yield of product was 75.0%.

Which mass of product was collected?

- A** 1.26 g **B** 1.98 g **C** 2.07 g **D** 2.70 g

- 17 Ammonium sulfate, $(\text{NH}_4)_2\text{SO}_4$, and ammonium nitrate, NH_4NO_3 , are used as fertilisers.

These salts have different percentages by mass of nitrogen. They have the same effect as each other on the pH of neutral soil.

Which row is correct?

	higher percentage of nitrogen by mass	effect on pH of soil
A	ammonium nitrate	decrease
B	ammonium nitrate	increase
C	ammonium sulfate	decrease
D	ammonium sulfate	increase

- 12** 1.15 g of a metallic element needs 300 cm^3 of oxygen for complete reaction, at 298 K and 1 atm pressure, to form an oxide which contains O^{2-} ions.

What could be the identity of this metallic element?

- A** calcium
- B** magnesium
- C** potassium
- D** sodium

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What could be the identity of this metallic element?

- A** calcium
- B** magnesium
- C** potassium
- D** sodium

3 *Use of the Data Booklet is relevant to this question.*

1.00 g of carbon is combusted in a limited supply of pure oxygen. 0.50 g of the carbon combusts to form CO_2 and 0.50 g of the carbon combusts to form CO.

The resultant mixture of CO_2 and CO is passed through excess NaOH(aq) and the remaining gas is then dried and collected.

What is the volume of the remaining gas? (All gas volumes are measured at 25°C and 1 atmosphere pressure.)

- A** 1 dm^3 **B** 1.5 dm^3 **C** 2 dm^3 **D** 3 dm^3

- 2 The shell of a chicken's egg makes up 5% of the mass of an average egg. An average egg has a mass of 50 g.

Assume the egg shell is pure calcium carbonate.

How many complete chicken's egg shells would be needed to neutralise 50 cm^3 of 2.0 mol dm^{-3} ethanoic acid?

- A** 1 **B** 2 **C** 3 **D** 4

- 9** *Use of the Data Booklet is relevant to this question.*

In an experiment, 12.0 dm^3 of oxygen, measured under room conditions, is used to burn completely 0.10 mol of propan-1-ol.

What is the final volume of gas, measured under room conditions?

- A** 7.20 dm^3 **B** 8.40 dm^3 **C** 16.8 dm^3 **D** 18.00 dm^3

- 6 Aluminium carbide, Al_4C_3 , reacts readily with aqueous sodium hydroxide. The two products of the reaction are $NaAlO_2$ and a hydrocarbon. Water molecules are also involved as reactants.

What is the formula of the hydrocarbon?

- A** CH_4 **B** C_2H_6 **C** C_3H_8 **D** C_6H_{12}

Question 26

9701_w14_qp_12 • Q6

- 6 Aluminium carbide, Al_4C_3 , reacts readily with aqueous sodium hydroxide. The two products of the reaction are $NaAlO_2$ and a hydrocarbon. Water molecules are also involved as reactants.

What is the formula of the hydrocarbon?

- A** CH_4 **B** C_2H_6 **C** C_3H_8 **D** C_6H_{12}

- 11 A solution of Sn^{2+} ions will reduce an acidified solution of MnO_4^- ions to Mn^{2+} ions. The Sn^{2+} ions are oxidised to Sn^{4+} ions in this reaction.

How many moles of Mn^{2+} ions are formed when a solution containing 9.5 g of SnCl_2 (M_r : 190) is added to an excess of acidified KMnO_4 solution?

- A** 0.010 **B** 0.020 **C** 0.050 **D** 0.125

17 *Use of the Data Booklet is relevant to this question.*

1.15 g of a metallic element reacts with 300 cm³ of oxygen at 298 K and 1 atm pressure, to form an oxide which contains O²⁻ ions.

What could be the identity of the metal?

- A** calcium
- B** magnesium
- C** potassium
- D** sodium

- 13** 0.02 mol of aluminium is burned in oxygen and the product is reacted with 2.00 mol dm^{-3} hydrochloric acid.

What minimum volume of acid will be required for complete reaction?

- A** 15 cm^3 **B** 20 cm^3 **C** 30 cm^3 **D** 60 cm^3

Answer Key

No.	Paper	Topic	Answer
1	9701_m23_qp_12	2. Atoms, molecules and stoichiomet...	B
2	9701_w23_qp_11	2. Atoms, molecules and stoichiomet...	A
3	9701_w23_qp_13	2. Atoms, molecules and stoichiomet...	A
4	9701_m22_qp_12	2. Atoms, molecules and stoichiomet...	D
5	9701_s22_qp_11	2. Atoms, molecules and stoichiomet...	A
6	9701_s22_qp_13	2. Atoms, molecules and stoichiomet...	D
7	9701_w22_qp_11	2. Atoms, molecules and stoichiomet...	A
8	9701_w22_qp_13	2. Atoms, molecules and stoichiomet...	A
9	9701_m21_qp_12	2. Atoms, molecules and stoichiomet...	A
10	9701_s21_qp_13	2. Atoms, molecules and stoichiomet...	D
11	9701_w20_qp_13	2. Atoms, molecules and stoichiomet...	C
12	9701_m19_qp_12	2. Atoms, molecules and stoichiomet...	A
13	9701_w19_qp_12	2. Atoms, molecules and stoichiomet...	A
14	9701_w19_qp_13	2. Atoms, molecules and stoichiomet...	C
15	9701_s18_qp_11	2. Atoms, molecules and stoichiomet...	C
16	9701_s17_qp_11	2. Atoms, molecules and stoichiomet...	C
17	9701_s17_qp_12	2. Atoms, molecules and stoichiomet...	B
18	9701_s17_qp_12	2. Atoms, molecules and stoichiomet...	D
19	9701_m16_qp_12	2. Atoms, molecules and stoichiomet...	A
20	9701_w16_qp_11	2. Atoms, molecules and stoichiomet...	D
21	9701_w16_qp_13	2. Atoms, molecules and stoichiomet...	D
22	9701_s15_qp_11	2. Atoms, molecules and stoichiomet...	A
23	9701_s15_qp_12	2. Atoms, molecules and stoichiomet...	B
24	9701_s14_qp_12	2. Atoms, molecules and stoichiomet...	B
25	9701_w14_qp_11	2. Atoms, molecules and stoichiomet...	A
26	9701_w14_qp_12	2. Atoms, molecules and stoichiomet...	A
27	9701_s13_qp_11	2. Atoms, molecules and stoichiomet...	B
28	9701_w12_qp_11	2. Atoms, molecules and stoichiomet...	D

Answer Key

No.	Paper	Topic	Answer
29	9701_s11_qp_11	2. Atoms, molecules and stoichiomet...	C